Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Driver 2 and matrix

File: matrix2.kicad_sch

Driver 3 and matrix

File: matrix3.kicad_sch

Driver 4 and matrix

File: matrix4.kicad_sch

Driver 5 and matrix

File: matrix5.kicad_sch

Rails: VBUS -> VSYS 3.0–4.5 V (LED SW lines) -> +3V3 -> +1V1 at the RP2350.
5x IS31FL3745-CLS4-TR WLCSP-36 drivers (47 RGB LEDs each = 235 total, zero exposed ground pads).
Nets between sheets are global labels; rails are power symbols.
Requirements: design/intent.md. Parts: design/parts/drivers-bga.md. Decisions: design/decisions.md.

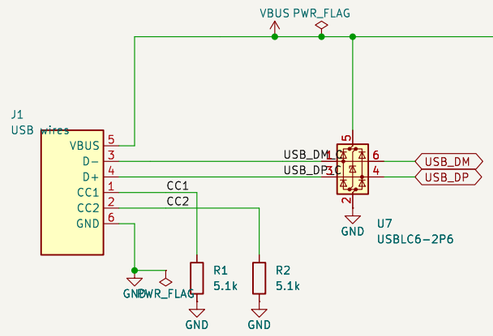
Drawn by scripts/bga/schematic_d.py
235 RGB LEDs on a 38 mm round board (5x IS31FL3745 WLCSP-36)
CERN-OHL-P-2.0

Sheet: /
File: round-pendant.kicad_sch

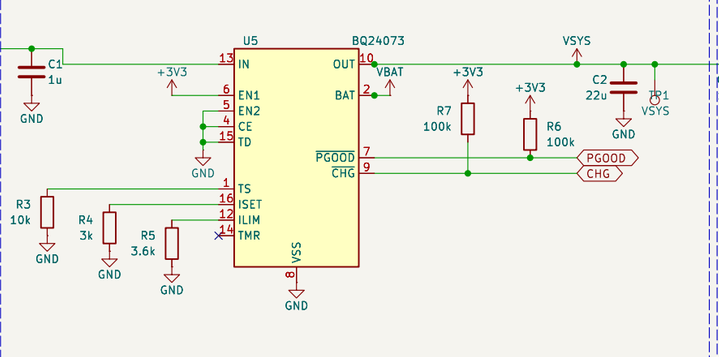
Title: round-pendant: top

Size: A4	Date: 2026-11-04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 1/8

USB WIRES, ESD

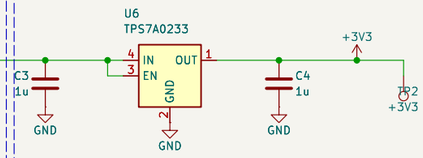


CHARGER AND POWER PATH



USB500: 450−500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

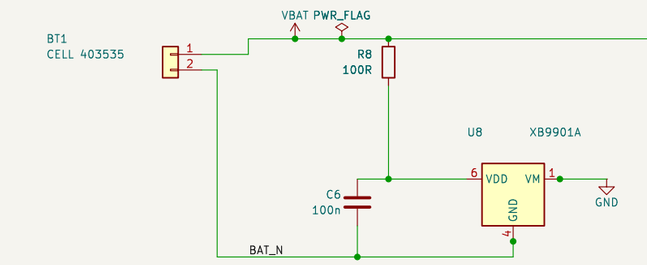
3V3



3V3 only ever follows VSYS down: the driver inputs
stay within VCC + 0.3 V (drivers−bga.md). 150 mA max.

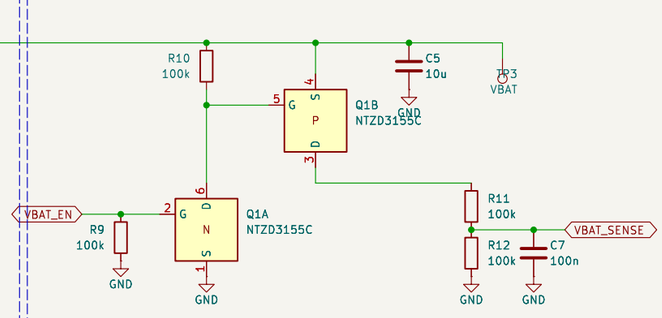
CELL AND PROTECTION

Hand−soldered leads. 403535 pouch cell, charged to 4.2 V (battery.md)



BAT_N: cell minus. Pins 4, 5 and the pad. Never pour GND here,
or protection is bypassed (protection.md).

VBAT SENSE



Zero draw off. ADC reads true only while 3V3 regulates.
VBAT >= 3.5 V (intent R22). Wait >= 5 ms after enabling.

Drawn by scripts/bga/schematic_d.py
USB−C wires, charger, protection, LDO, VBAT sense
CERN−OHL−P−2.0

Sheet: /Power/
File: power.kicad_sch

Title: round−pendant: power

Size: A3	Date: 2026−11−04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 2/8

RP2350A, FLASH, CRYSTAL

Regulator: C16 CIN, C18 COUT, L1's polarity and copper follow chips/rp2350.md

100 nF at IOVDD pins 1, 11, 20, 30, 38 (C8–C12), USB_OTP_VDD 53 and QSPI_IOVDD 54 (C13), ADC_AVDD 44 and IOVDD 45 (C14).

C18 COUT at L1; C19 at DVDD 23; C20, C21 at DVDD 6, 39

+3V3
C8 100n
C9 100n
C10 100n
C11 100n
C12 100n
C13 100n
C14 100n
GND
R13 33R
VREG_AVDD
PWR_FLAG
C17 4.7u
L1 3.3u
VREG_LX
C16 4.7u
C18 4.7u
C19 4.7u
C20 100n
C21 100n
PWR_FLAG +1V1
TP4 1V1

U3 RP2350A

Pin	Net
46	VREG_AVDD
53	USB_OTP_VDD
44	ADC_AVDD
54	QSPI_IOVDD
1	IOVDD
49	VREG_VIN
48	VREG_LX
50	VREG_FB
6	DVDD
26	RUN
51	USB_DM
52	USB_DP
60	QSPI_SS
56	QSPI_SCLK
57	QSPI_SD0
59	QSPI_SD1
58	QSPI_SD2
55	QSPI_SD3
21	XIN
22	XOUT
24	SWCLK
25	SWDIO
47	VREG_PGND
61	GND
2	GPIO0
3	GPIO1
4	GPIO2
5	GPIO3
7	GPIO4
8	GPIO5
9	GPIO6
10	GPIO7
12	GPIO8
13	GPIO9
14	GPIO10
15	GPIO11
16	GPIO12
17	GPIO13
18	GPIO14
19	GPIO15
27	GPIO16
28	GPIO17
29	GPIO18
31	GPIO19
32	GPIO20
33	GPIO21
34	GPIO22
35	GPIO23
36	GPIO24
37	GPIO25
40	GPIO26/ADC0
41	GPIO27/ADC1
42	GPIO28/ADC2
43	GPIO29/ADC3

ACC_SDA
ACC_SCL
I2C_SDA
I2C_SCL
LED_SDB
ACC_INT
BTN
CHG
PGOOD
VBAT_EN
VBAT_SENSE

R14 27R
R15 27R
USB_DM
USB_DP
USB_DM_MCU
USB_DP_MCU

U4 W25Q128JVPIQ
C22 100n
+3V3
GND
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
GND

Y1 12MHz
C23 15p
C24 15p
R17 1k

Flash pad: float (no internal connection)

VREG_PGND joins GND at one point

I2C0 (5x IS31FL3745, 1 MHz): 12 mA drive, LGC=1 (drivers-bga.md).
LED_SDB blanks all 5 (~5 uA sleep).

ACCELEROMETER (WAKE)

+3V3
+3V3+3V3
R20 4.7k
R19 4.7k
C25 100n
C26 4.7u
C27 100n
GND
ACC_SCL
ACC_SDA
ACC_INT
U9 LIS2DW12
SCL
SDA
CS
SAO
VDD
VDD_IO
INT1
INT2
GND
NC

Address 0x19. CS, SAO tied high: ~20 k internal pull-ups would draw 160 uA if strapped low.

BUTTONS

R18 1k
BTN
SW1 USER / WAKE
GND

Internal pull-up; wake on low. Hold SW2, tap SW1 for BOOTSEL.

QSPI_SS
R16 1k
SW2 BOOTSEL
GND

I2C0 PULL-UPS & LED_SDB

+3V3
R21 1k
I2C_SDA
R22 1k
I2C_SCL
LED_SDB
R31 100k
C35 100n
GND

TEST PADS

TP5 RUN
TP8 GND
TP6 SWCLK
TP7 SWDIO

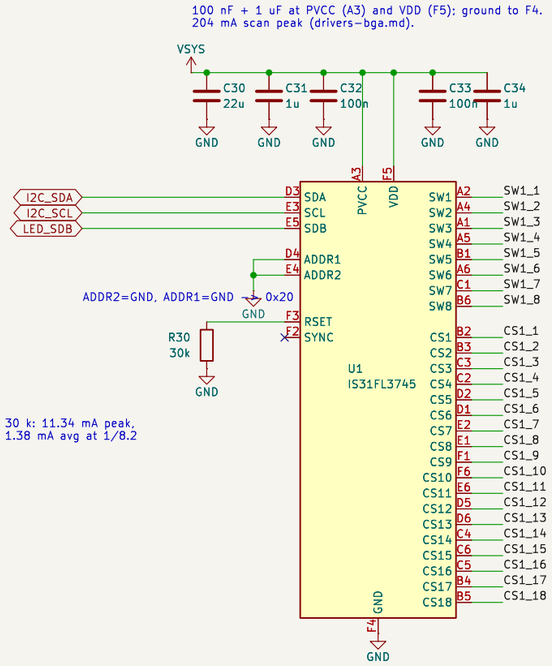
Drawn by scripts/bga/schematic_d.py
RP2350A, flash, crystal, accelerometer, buttons, I2C0 & LED_SDB
CERN-OHL-P-2.0

Sheet: /MCU/
File: mcu.kicad_sch

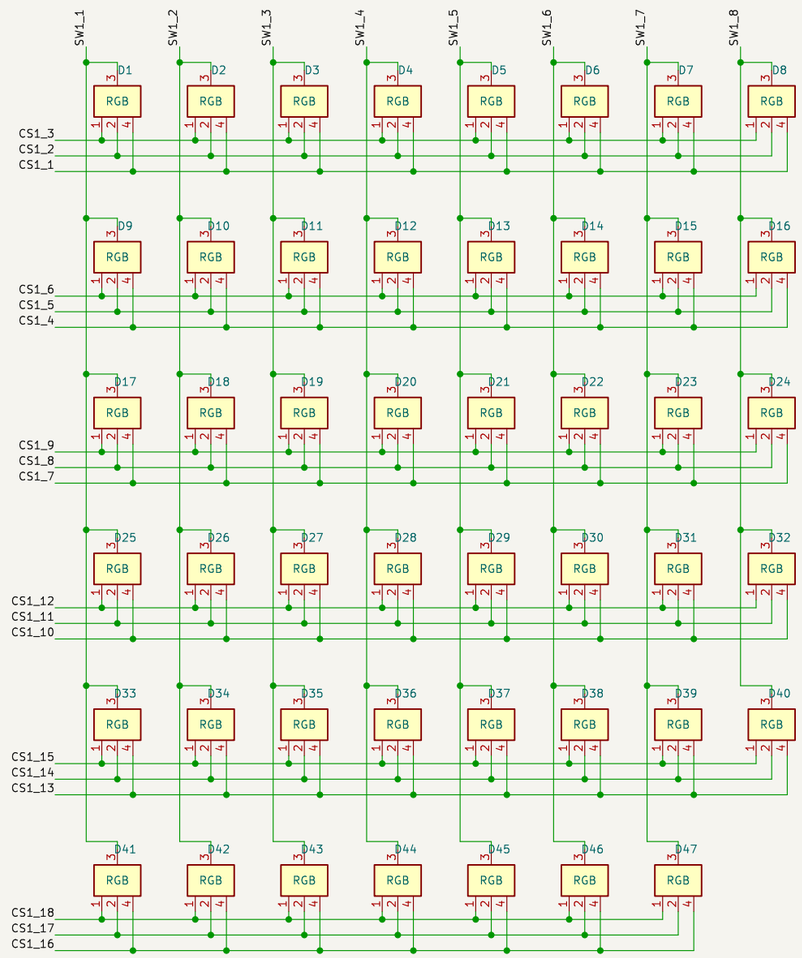
Title: round-pendant: mcu

Size: A3	Date: 2026-11-04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 3/8

DRIVER 1 (U1, 0x20)



47 RGB LEDS (6 x 8 − 1): R = CS(3k), G = CS(3k−1), B = CS(3k−2)



Drawn by scripts/bga/schematic_d.py
U1 IS31FL3745-CLS4-TR (0x20), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

Title: round-pendant: driver 1 and matrix

Size: A3	Date: 2026-11-04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 4/8

DRIVER 2 (U2, 0x22)

100 nF + 1 uF at PVCC (A3) and VDD (F5); ground to F4.
204 mA scan peak (drivers-bga.md).

VSYS

C40 22u — C41 1u — C42 100n — C43 100n — C44 1u — GND

I2C_SDA
I2C_SCL
LED_SDB

U2 IS31FL3745

D3	SDA	SW1	A2	SW2_1
E3	SCL	SW2	A4	SW2_2
E5	SDB	SW3	A1	SW2_3
		SW4	A5	SW2_4
D4	ADDR1	SW5	B1	SW2_5
E4	ADDR2	SW6	A6	SW2_6
		SW7	C1	SW2_7
		SW8	B6	SW2_8
F3	RSET			
F2	SYNC	CS1	B2	CS2_1
		CS2	B3	CS2_2
		CS3	C3	CS2_3
		CS4	C2	CS2_4
		CS5	D2	CS2_5
		CS6	D1	CS2_6
		CS7	E2	CS2_7
		CS8	E1	CS2_8
		CS9	F1	CS2_9
		CS10	F6	CS2_10
		CS11	E6	CS2_11
		CS12	D5	CS2_12
		CS13	D6	CS2_13
		CS14	C4	CS2_14
		CS15	C6	CS2_15
		CS16	C5	CS2_16
		CS17	B4	CS2_17
		CS18	B5	CS2_18
A3	PVCC			
F5	VDD			
F4	GND			

ADDR2=GND, ADDR1=SDA → 0x22

R33 30k — GND

30 k: 11.34 mA peak,
1.38 mA avg at 1/8.2

47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)

	SW2_1	SW2_2	SW2_3	SW2_4	SW2_5	SW2_6	SW2_7	SW2_8
CS2_3	D48	D49	D50	D51	D52	D53	D54	D55
CS2_2								
CS2_1								
CS2_6	D56	D57	D58	D59	D60	D61	D62	D63
CS2_5								
CS2_4								
CS2_9	D64	D65	D66	D67	D68	D69	D70	D71
CS2_8								
CS2_7								
CS2_12	D72	D73	D74	D75	D76	D77	D78	D79
CS2_11								
CS2_10								
CS2_15	D80	D81	D82	D83	D84	D85	D86	D87
CS2_14								
CS2_13								
CS2_18	D88	D89	D90	D91	D92	D93	D94	
CS2_17								
CS2_16								

Each: RGB (1=R, 2=G, 4=B; 3=common)

Drawn by scripts/bga/schematic_d.py
U2 IS31FL3745-CLS4-TR (0x22), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

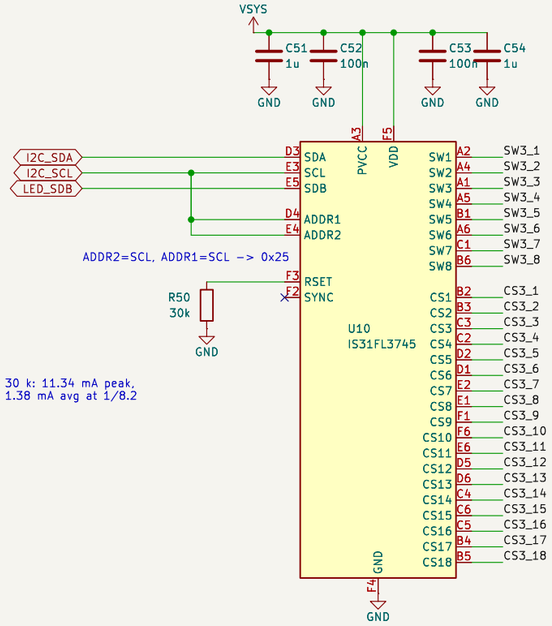
Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

Title: round-pendant: driver 2 and matrix

Size: A3	Date: 2026-11-04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 5/8

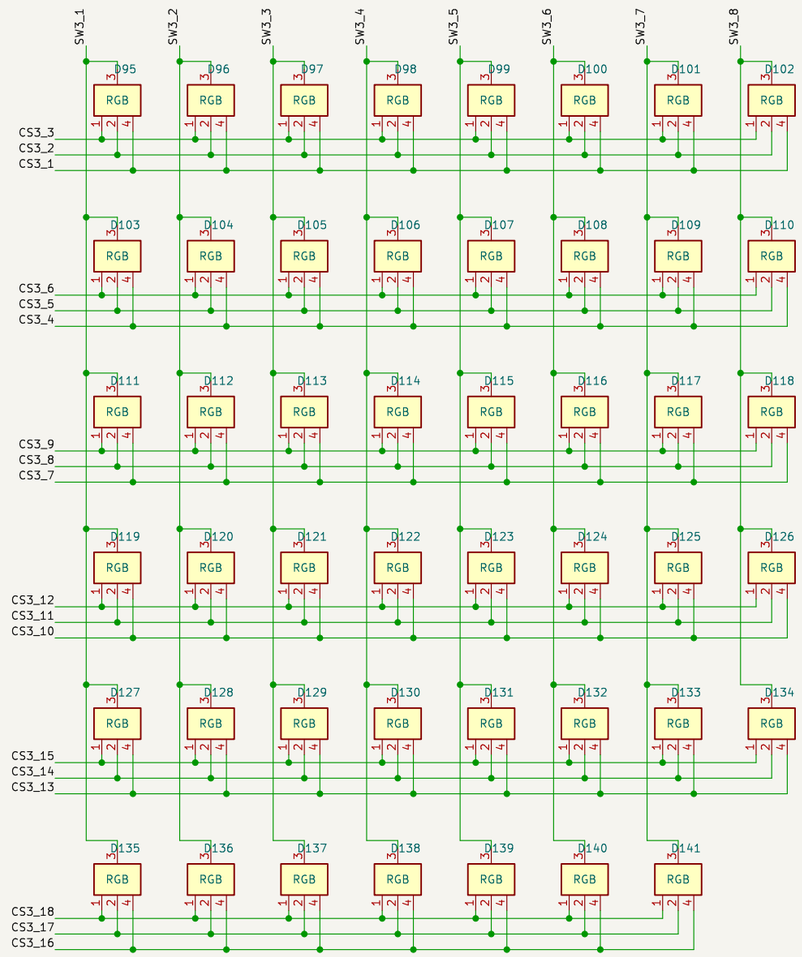
DRIVER 3 (U10, 0x25)

100 nF + 1 uF at PVCC (A3) and VDD (F5); ground to F4.
204 mA scan peak (drivers-bga.md).



30 k: 11.34 mA peak,
1.38 mA avg at 1/8.2

47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)



Drawn by scripts/bga/schematic_d.py
U10 IS31FL3745-CLS4-TR (0x25), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

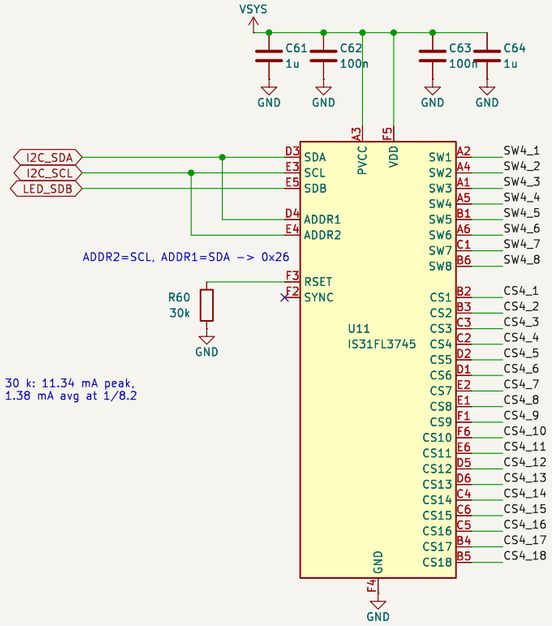
Sheet: /Driver 3 and matrix/
File: matrix3.kicad_sch

Title: round-pendant: driver 3 and matrix

Size: A3	Date: 2026-11-04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 6/8

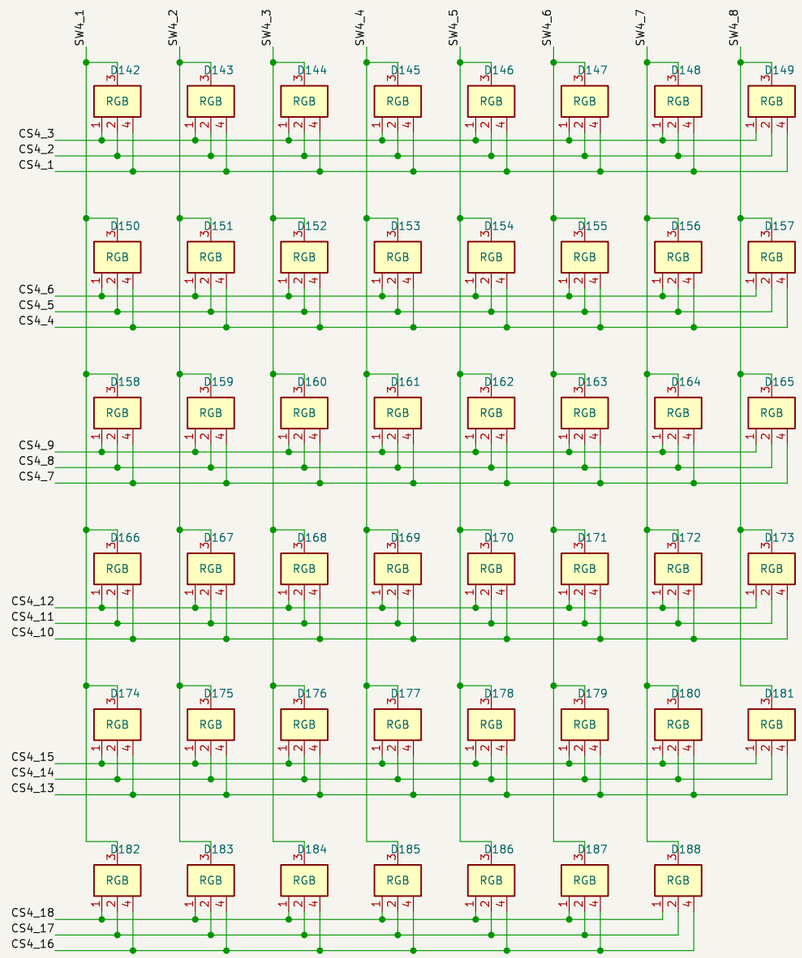
DRIVER 4 (U11, 0x26)

100 nF + 1 uF at PVCC (A3) and VDD (F5); ground to F4.
204 mA scan peak (drivers-bga.md).



30 k: 11.34 mA peak,
1.38 mA avg at 1/8.2

47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)



Drawn by scripts/bga/schematic_d.py
U11 IS31FL3745-CLS4-TR (0x26), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

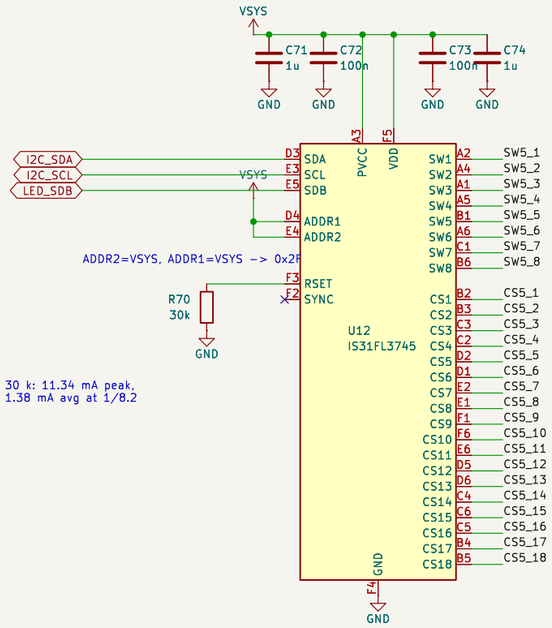
Sheet: /Driver 4 and matrix/
File: matrix4.kicad_sch

Title: round-pendant: driver 4 and matrix

Size: A3	Date: 2026-11-04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 7/8

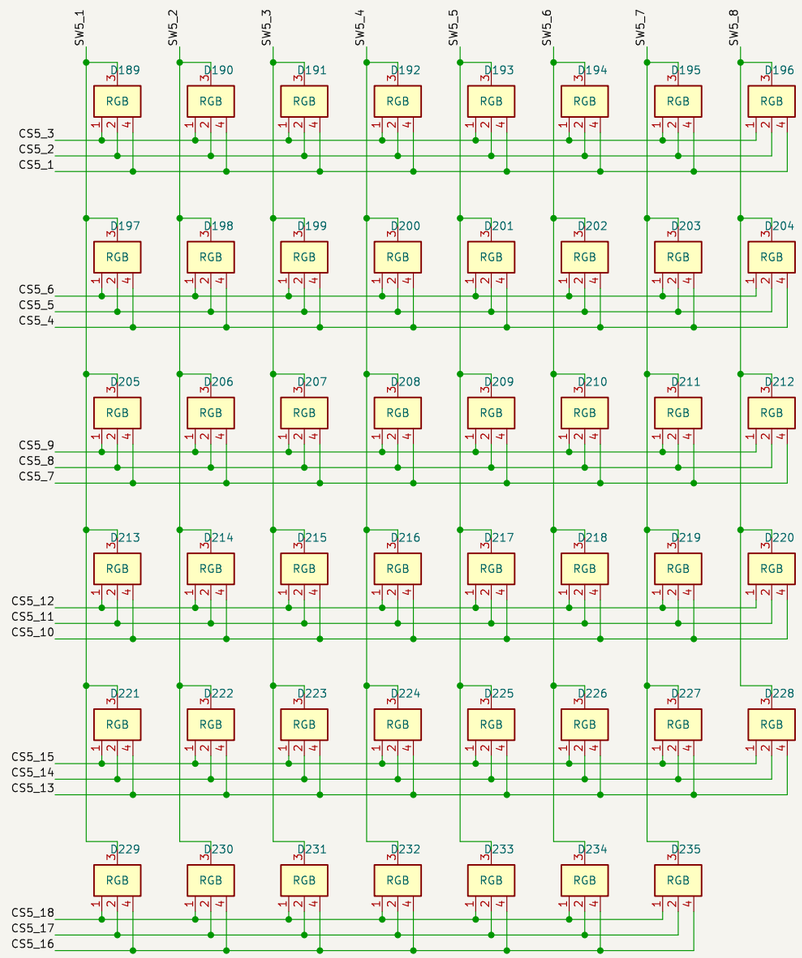
DRIVER 5 (U12, 0x2F)

100 nF + 1 uF at PVCC (A3) and VDD (F5); ground to F4.
204 mA scan peak (drivers-bga.md).



30 k: 11.34 mA peak,
1.38 mA avg at 1/8.2

47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)



Drawn by scripts/bga/schematic_d.py
U12 IS31FL3745-CLS4-TR (0x2F), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

Sheet: /Driver 5 and matrix/
File: matrix5.kicad_sch

Title: round-pendant: driver 5 and matrix

Size: A3	Date: 2026-11-04	**Rev: D**
KiCad E.D.A. 9.0.8		Id: 8/8